20 September 1973
SUPERSEDING
MIL-F-10805C
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MILITARY SPECIFICATION

FUEL, COMPRESSED, TRIOXANE, RATION HEATING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers a compressed chemical fuel with low luminosity characteristics (see 6.1).

1.2 Classification. The compressed fuel shall be of the following types as specified (see 6.2):

Type I - 30 grams (scored).
Type II - 15 grams (unscored).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

FSC 9110

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The ingredients of the compressed fuel shall be as follows:

3.1.1 Trioxane. Trioxane shall be of such purity that the performance of finished fuel bars compressed therefrom, using the formulation in table I, shall meet the requirements of this specification.

3.1.2 Magnesium stearate. Magnesium stearate shall be a low-luminosity powder of such fineness that 100 percent will pass through a 325-mesh sieve (see table II).

3.1.3 Dye. The dye used to impart color to the fuel shall be methylene blue (see table II).

3.2 Chemical composition. The chemical composition of the compressed fuel shall conform to the requirements shown in table I when tested as specified in table II.

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TABLE I. Chemical composition

	Percent by weight Minimum	Maximum
Trioxane	95.0	----
Magnesium Stearate (calculated as residue after burning).	----	5.0
Dye	As required	

3.3 Form.

3.3.1 Type I. The compressed fuel shall be a densely molded, light blue, non-brittle bar of uniform texture. Each bar shall be 3-1/4 ± 1/16 inches long, 1-1/4 ± 1/16 inches wide and 3/8 ± 1/16 inch in thickness. One face of the bar shall be indented in two places to a depth of 1/32 inch to facilitate breaking into three approximately equal pieces when desired. Corners of the bar shall be rounded and the edges beveled or rounded.

3.3.2 Type II. The compressed fuel shall be a densely molded, light blue, non-brittle bar of uniform texture. Each bar shall be 1-5/8 ± 1/16 inches long, 1-1/4 ± 1/16 inches wide and 3/8 ± 1/16 inch in thickness. The bar shall not be indented. Corners of the bar shall be rounded and the edges beveled or rounded.

3.4 Weight.

3.4.1 Type I. The finished bar shall weigh 30 ± 1 grams.

3.4.2 Type II. The finished bar shall weigh 15 ± 0.5 grams.

3.5 Performance. When tested as specified in 4.4.5, the compressed fuel shall conform to the following requirements:

3.5.1 Stability. Fuel shall maintain its solid state while burning.

3.5.2 Odor of combustion products. There shall be no strong odor of formaldehyde.

3.5.3 Flame characteristics. The fuel shall burn with a steady, blue flame of low luminosity with only occasional flashes of color other than blue and with no objectionable increase in luminosity.

3.5.4 Fuel life.

3.5.4.1 Type I. At ambient temperatures between 65° and 80°F. the burning time for one bar of fuel shall be not less than six (6) minutes.

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3.5.4.2 Type II. At ambient temperatures between 65° and 80°F. the burning time for one bar of fuel shall be not less than five (5) minutes.

3.6 Labeling.

3.6.1 Unit package.

3.6.1.1 Type I. One side of each wrapper on each individual bar shall be legibly printed with black water-resistant ink conforming to the test specified in 4.4.7 with the following information:

FUEL, COMPRESSED, TRIOXANE
RATION HEATING
SPECIFICATION:
FOR HEATING COMPONENTS OF A MEAL
DIRECTIONS:

WARNING: HARMFUL IF SWALLOWED.
Take precautions to avoid contamination of food by loose powder. Cleanse hands after handling.

Do not remove wrapper until ready to use.
Remove any loose powder from tablet before igniting.
Place and burn 1/3 bar or more approximately 2 inches below vessel being heated.
Protect from wind but use with adequate ventilation.
Replace partially unused bar in tightly closed wrapper.
Name and address of manufacturer.

3.6.1.2 Type II. One side of each wrapper shall be legibly printed with black, water-resistant ink conforming to the test specified in 4.4.7 with the following information:

FUEL, COMPRESSED, TRIOXANE
RATION HEATING
SPECIFICATION:
FOR HEATING COMPONENTS OF A MEAL

On the reverse side of each wrapper, add the following information:

DIRECTIONS

WARNING: HARMFUL IF SWALLOWED.
Take precautions to avoid contamination of food by loose powder. Cleanse hands after handling.

Do not remove wrapper until ready to use.
Remove any loose powder from tablet before igniting.

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Place and burn entire bar approximately 2 inches below vessel being heated.
Protect from wind but use with adequate ventilation.
Name and address of manufacturer.

3.7 Workmanship. The fuel bars shall be densely molded, uniform in composition and texture with no separation of components and shall not be brittle. The blue color shall be reasonably uniformly distributed throughout the bar. Score marks, where applicable, shall be sharp and straight. Wrappers shall fit the bars snugly and show no leakage when tested in accordance with 4.4.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and tests. Sampling for inspection and tests shall be in accordance with the provisions of Standard MIL-STD-105, except where otherwise indicated. For purposes of sampling for inspection, a lot shall consist of all material of the same type and contruction from the same manufactured batch and shall be limited to the number of bars produced in a single workday (24 hours).

4.2.1 Inspection of components and materials. The supplier shall be responsible for insuring that the quality assurance provisions for components or materials shall be in accordance with this specification and with referenced subsidiary specifications to the extent applicable. In the event of conflict, this specification shall govern.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined in accordance with the classification of defects in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) set forth in 4.3.1.4. Random samples shall be drawn from each lot of end items of the same type for examination of visual, contents and preparation for delivery defects. The lot size for purposes of determining the sample size,

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in accordance with Standard MIL-STD-105, shall be expressed in units of filled primary containers of the same capacity, consisting of compressed fuel manufactured in one batch for the examination in 4.3.1.1 and in units of shipping containers for examination under 4.3.1.2.

4.3.1.1 Examination of the packaged end item for defects in form, color, workmanship, packaging and marking. The sample unit for this examination shall be one packaged fuel bar.

Examine	Defect
Exterior appearance of packaged fuel bar for:	
Form	Any evidence of improper or incomplete seal (examine for visible channels). Not densely molded bar. Not uniform texture. Corners not rounded. Edges not beveled or rounded. One face not indented in two places (type I only). Indentations do not divide bar in approximately 3 equal parts (type I only).
Color	Not light blue. Not reasonably uniform.
Workmanship	Crumbly (brittle). Separation of components; magnesium stearate particles separated from bar. Surface not smooth. Any broken edge. Score marks not straight and sharp (type I only).
Container (individual wrap):	
Construction	Not as specified (see 5.1).
Seals	Not sealed on four sides. Any evidence of dust or other foreign matter on inside of seals (see 6.4). Less than 3/8 inch in width. Fins bent or broken.
Color (exterior)	Not OD. Not uniform.

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Examine	Defect
Workmanship	Any dirt or foreign matter. Any cut, tear or puncture. Fuel bar not centered.
Markings	Missing, incomplete, illegible, incorrect (see 3.6).

4.3.1.2 Examination of preparation for delivery. An examination shall be made to determine that markings, materials, workmanship and contents comply with the requirements of Section 5 of this specification. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers, fully prepared for delivery, shall be examined for closure defects.

Examine	Defect
Markings (packages and shipping containers)	Incorrect, incomplete, illegible, omitted; of improper size, sequence or method of application; not in accordance with contract requirements.
Material	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components such as: incomplete closure of container flaps, loose or inadequate sealing, strapping or stapling. Bulged or distorted containers.
Contents	Number of unit packages per shipping container is more or less than required. Intermediate package not as specified (see 5.1.1.2). Number of unit packages per intermediate package is less than specified.

4.3.1.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels and the acceptable quality levels (AQLs) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1.1	I	1.5
4.3.1.2	S-2	4.0

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4.3.2 Testing of the end item. The end item shall be tested for the characteristics specified in Section 3 in accordance with test methods specified in table II. The lot size for test purposes shall be the number of containers of the same type containing compressed fuel of the same type from a single manufactured batch. The sample unit for all tests except wrapper leakage shall be 14 packaged bars. The sample size for all tests except the water leakage test shall be 3. For the water leakage test the sample unit shall be three packaged bars and the sample size shall be the same as used for examination in 4.3.1.1 (the same samples may be used except that the water leakage test should be performed first). A batch is defined as the amount of product made from the same materials, at the same time and under the same manufacturing process. A single determination shall be made on each sample unit for all characteristics. The lot shall be unacceptable if there is any failure to meet any characteristic applicable to the sample unit.

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INSTRUCTIONS FOR TESTING OF END ITEM

TABLE II

CHARACTERISTIC	Specification Reference		Requirements Applicable To		Number Determinations	Results Reported As		Inspect Level	AQL
	Requirement	Test Method	Individ Unit	Lot Aver	Per Unit	Pass or Fail	Numerically to Nearest		
Components:									
Magnesium stearate, luminosity and fineness.	3.1.2	(1/)							
Dye, methylene blue	3.1.3	(1/)							
Chemical composition:									
Trioxane, percent	Table I	4.4.1	X		1	X	1.0		
Magnesium stearate, percent	Table I	4.4.2	X		1	X	1.0		
Weight:									
Type I, grams	3.4.1	4.4.3	X		1 (average of 10 bars)	X	1.0		
Type II, grams	3.4.2	4.4.3	X		1 (average of 10 bars)	X	0.5		
Form:									
Type I:									
Length, width and thickness, inches.	3.3.1	4.4.4	X		1 (average of 3 bars).	X	1/16		
Depth of scoring	3.3.1	4.4.4	X		1 (average of 3 bars)	X	1/32		
Type II:									
Length, width, and thickness, inches.	3.3.2	4.4.4	X		1 (average of 3 bars)	X	1/16		
Performance:									
Stability	3.5.1	4.4.5	X		1	X			
Odor	3.5.2	4.4.5	X		1	X			
Flame characteristics	3.5.3	4.4.5	X		1	X			
Fuel life, minutes	3.5.4	4.4.5	X		1	X	1.0		
Wrapper leakage	3.7	4.4.6	X		1	X			
Water resistance of ink	3.6	4.4.7	X		1	X			

1/ The supplier shall submit to the Contracting Officer a certif cate of compliance for these characteristics.

2/ If failure is indicated, report description or numerical point of failure, as applicable.

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4.4 Test procedures.

4.4.1 Trioxane. A 10.0 gram sample of the fuel bar shall be exposed in air at 100 ± 5°F. for 96 hours and then reweighed. The loss in weight expressed in percent is trioxane.

4.4.2 Magnesium stearate. Weigh one fuel bar to the nearest 0.1 gram. Place bar in a tared aluminum dish and burn in an atmosphere free from air currents. After burning is completed weigh the ash and calculate the percentage of magnesium stearate as follows:

4.4.2.1 Calculation.

$$\text{Percent magnesium stearate} = \frac{\text{Weight of residue} \times 100}{\text{weight of bar}}$$

4.4.3 Weight. The weight of the finished bar shall be determined by individually weighing ten (10) bars and calculating the average weight per bar.

4.4.4 Form. Length, width and thickness and depth of scoring in type I bar, shall be determined by individually measuring three (3) bars and calculating the average dimensions per bar.

4.4.5 Performance. Ignite one bar at room temperature (72°F ± 5°F.) in an atmosphere free from air currents. Note the time of ignition. During burning, observe the following: (a) Steadiness of flame, (b) color and luminosity of flame, (c) odor of volatile products of combustion, (d) condition of unburned portion prior to complete combustion whether solid or liquid. Note time when burning ceases and calculate fuel life in minutes.

4.4.6 Leakage in bar wrapper. Place one wrapped bar in a desiccator partially filled with water. Cover the bar with a grating to prevent floating and to maintain at least a one inch height of water over the top surface of the packaged bar. Gradually apply a vacuum of 10 inches of mercury to the air space in the desiccator for 30 seconds. Observe the packaged bar for a steady flow of air bubbles (leakage indication) during this period. Remove the wrapped bar from the desiccator, dry hands and wrapped bar, and then open the package and examine interior of wrapper and bar for presence of moisture.

4.4.7 Water-resistance of ink. Wrapper with marking shall be immersed in water for a period of one hour with water maintained at a temperature of 80 ± 2°F. Upon removal from the water the markings shall be examined for smearing and legibility.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

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5.1.1 Level A.

5.1.1.1 Unit package. Each fuel bar shall be packaged in a sheet of material made from aluminum 0.002-inch minimum thick, coated inside with a 0.001-inch polyethylene coating. Alternatively, the package material may be made from dead soft aluminum 0.0016 inch minimum thickness, coated inside with a 0.0015-inch polyethylene coating. The package shall be formed by placing the bar between two sheets of the material specified and heat sealing edges on all four sides (see 6.4). Heat seals shall be not less than three-eights of an inch in width and shall not be bent or otherwise distorted after sealing. The exterior surface of the finished package shall be olive drab in color. The finished package shall be subjected to the leakage test specified in 4.3.2.

5.1.1.2 Intermediate package. Three fuel bars of one type only, unit packaged as specified in 5.1.1.1, shall be packaged in a snug-fitting folding paperboard box conforming to variety 1, style II, type D, class a of Specification PPP-B-566. The exterior surface of the box shall be a dull non-reflecting color as specified in the box specification. The box shall fit the bars in such a manner that movement during transit and consequent bending and cracking of the heat seals will be prevented. The bars shall be packaged flat within the box, stacked one on top of another. Caution shall be taken at time of inserting bars into the box so that the seals will not be bent or flexed.

5.1.2 Level C. Fuel bars shall be packaged to afford adequate protection against physical damage or deterioration during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Seven hundred and fifty type I fuel bars or twelve hundred type II fuel bars, packaged as specified in 5.1, shall be packed flat in a snug-fitting fiberboard shipping container conforming to style FOL, grade V2s of PPP-B-636. Each shipping container shall be provided with full size top and bottom fiberboard pads fabricated of the same material as the container. Each fiberboard container shall be closed method III, waterproofed method V, and reinforced in accordance with the appendix of PPP-B-636.

5.2.2 Level B. Seven hundred and fifty type I fuel bars or twelve hundred type II fuel bars, packaged as specified in 5.1, shall be packed flat in a snug-fitting fiberboard shipping container conforming to style FOL, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be provided with full size top and bottom fiberboard pads fabricated of the same material as the container. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

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5.2.2.1 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed method III in accordance with the appendix of the container specification.

5.2.3 Level C. Fuel bars, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. Each unit package shall be labeled as specified in 3.6.

5.3.1 Labeling of intermediate package.

5.3.1.1 Type I. Each paperboard box shall be legibly printed on one major panel with black water-resistant ink conforming to the test specified in table II with the following information:

FUEL, COMPRESSED, TRIOXANE,
RATION HEATING
Federal Stock number:
3 BARS
Specification:
DIRECTIONS (As specified for unit package, type I, see 3.6.1.1)
Name and address of manufacturer

5.3.1.2 Type II. Each paperboard box shall be legibly printed on one major panel with black water-resistant ink conforming to the test specified in table II with the following information:

FUEL, COMPRESSED, TRIOXANE,
RATION HEATING
Federal Stock number:
3 BARS
Specification:
DIRECTIONS (As specified for unit package, type II, see 3.6.1.2)
Name and address of manufacturer

5.3.2 Shipping container. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL-STD-129. Each shipping container shall also be marked with the following:

"STORE IN COOL DRY PLACE"

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6. NOTES

6.1 Intended use. Fuel Compressed Trioxane is intended for the emergency heating of field rations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Characteristics of trioxane fuel bar. The following properties characterize the trioxane fuel bar covered by this specification:

(a) Lights readily at both high and low temperatures.
(b) Burns with a blue flame of low luminosity.
(c) Flame is resistant to wind.
(d) Because of volatility of trioxane at ordinary temperatures bar must be kept hermetically sealed to pervent loss of evaporation.

6.4 Proper packaging procedures. Before commencing the heat sealing operation, the following procedures should be practiced for proper sealing:

(1) The inner coating on the thermoplastic film should be free from powder and dust.
(2) The tablet should be properly centered on the automatic heat sealer.
(3) The crimping sealer should be sufficiently heated before sealing the edges of the package.

Custodians:

Army - GL
Air Force - 68

Review activities:

Army - MD

User activity:

Navy - YD, MC

Preparing activity:

Army - GL

Project No. 9110-0010

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DOCUMENT IDENTIFIER AND TITLE

Fuel, Compressed, Trioxane, Ration Heating MIL-F-10805D

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A. GIVE PARAGRAPH NUMBER AND WORDING

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